

DataCo Supply Chain Analytics & Predictive ML

Business Analytics and Predictive Risk Case Study

\$36.78M

SALES

54.83%

LATE DELIVERY RATE

74.2%

ROC-AUC

\$3.97M

PROFIT

10.78%

PROFIT MARGIN

\$153,080

ESTIMATED SCENARIO NET SAVINGS

Business question

Where is delivery risk concentrated, what commercial patterns matter, and how can high-risk orders be identified before shipment execution?

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Dataset: DataCo Supply Chain Dataset | Evidence: final project package and executed outputs

01 Executive brief

A concise answer to the business problem, analytical response, evidence, and modeled value.

The analysis examines a 180,519-row supply chain dataset to connect delivery execution, commercial performance, predictive risk, and financial scenario impact in one decision-oriented workflow.

The operating baseline is material: 54.83% of order items are classified as late delivery. Across the cleaned dataset, sales are \$36.78M, profit is \$3.97M, and the resulting profit margin is 10.78%.

Operational analysis shows a clear difference between shipping modes. First Class has an observed late-delivery rate of 95.32%, compared with 38.07% for Standard Class. The overall average is 2.93 scheduled shipping days versus 3.50 actual shipping days.

The predictive layer uses nine order-time features and a leakage-controlled Gradient Boosting pipeline. On the holdout evaluation, the model records 84.3% precision, 55.0% recall, 66.5% F1, and 74.2% ROC-AUC, with 69.5% 5-fold cross-validation accuracy.

The deployment layer reuses the saved preprocessing and model pipeline in Streamlit. An example high-risk order produces a 97.0% late-delivery probability and a probability-weighted \$19.41 exposure under the \$20.00 scenario penalty.

The notebook scenario estimates \$153,080 net scenario savings and 38.66% scenario ROI across the 36,104 test orders using the model's binary holdout predictions. These figures are assumption-based scenario estimates, not realized financial savings.

The decision chain

Measure exposure → Locate concentration → Protect commercial value → Screen pre-shipment risk → Prioritize intervention → Evaluate scenario economics

Verified project scale

180,519 ORDER ITEMS	\$36.78M SALES	\$3.97M PROFIT
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02 Business problem and analytical objective

The project is structured around the questions that determine operational and commercial action.

The primary problem is not simply that late deliveries occur. The material issue is the combination of delivery exposure, uneven operational risk, commercial value at stake, and limited ability to identify higher-risk orders before shipment execution.

The analytical objective is therefore to move from descriptive visibility to a practical screening workflow: quantify the exposure, identify where it is concentrated, understand the commercial context, predict risk using information available at order placement, and connect the prediction to transparent intervention economics.

Questions answered in this report

QUESTION	ANSWER
What is the scale of the problem?	54.83% late-delivery exposure across 180,519 order items.
Where does operational risk concentrate?	Shipping-mode differences are substantial; the Power BI risk view also identifies regional concentration.
Where is commercial value concentrated?	Europe is the highest-sales market, Fishing is the leading sales category, and Consumer is the largest customer segment.
Can risk be screened before shipment?	Yes. The model uses order-time inputs and excludes post-shipment delivery outcomes.
How reliable is the screening model?	84.3% precision, 55.0% recall, 66.5% F1, 74.2% ROC-AUC, and 69.5% cross-validation accuracy.
How does the model become operational?	The saved pipeline is used by Streamlit to return probability, risk band, recommended action, and scenario exposure.
What is the modeled financial opportunity?	\$153,080 estimated net scenario savings and 38.66% scenario ROI under explicit assumptions.
Are the financial results realized savings?	No. They are scenario estimates that require operational validation.

Analytical objectives

- Quantify delivery exposure and establish a reliable operating baseline.
- Identify operational concentration by shipping mode and geography.
- Connect service performance with sales, profit, segment, category, and discount behavior.
- Develop a leakage-controlled pre-shipment late-delivery classifier.
- Convert model probability into an operational review signal.
- Estimate financial impact under explicit, auditable scenario assumptions.

03 Big Data and analytical framework

The analytical boundary is explicit so that descriptive findings and predictive claims remain distinct.

The project uses the public [DataCo Supply Chain on Kaggle](#) as the data source and analytical baseline. The saved cleaned dataset contains 180,519 rows and 46 columns. Two analytical fields are subsequently added in the notebook, resulting in a final analytical dataframe of 48 columns. The supplied field-description file defines the business meaning of delivery status, shipping duration, market, customer segment, sales, discounts, profit, order region, and other operational fields.

The analysis separates retrospective business evidence from forward-looking prediction. Historical delivery rates are used to understand operational patterns. The predictive model is restricted to information that would be available at order placement.

Predictive feature boundary: MODEL INPUT	BUSINESS MEANING
Type	Transaction type
Days for shipment (scheduled)	Scheduled delivery time
Shipping Mode	Selected shipping service
Order Item Quantity	Quantity on the order item
Sales	Sales value
Order Item Discount	Discount value
Category Name	Product category
Department Name	Product department
Order Region	Destination region

Leakage control

Post-shipment outcome fields are excluded from the predictive feature set. The prediction point is order placement.

Analytical flow

Data: Cleaned analytical frame	Analysis: Operational and commercial evidence	Prediction: Gradient Boosting probability	Action: Risk band and review recommendation	Economics: Scenario-based financial context
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04 Executive performance

The dashboard establishes the commercial baseline and the scale of delivery exposure.

The business generates substantial sales and profit, but delivery reliability represents a material operating exposure. The executive view places sales, profit, margin, order volume, and late delivery in the same frame so that service performance is evaluated alongside financial performance.

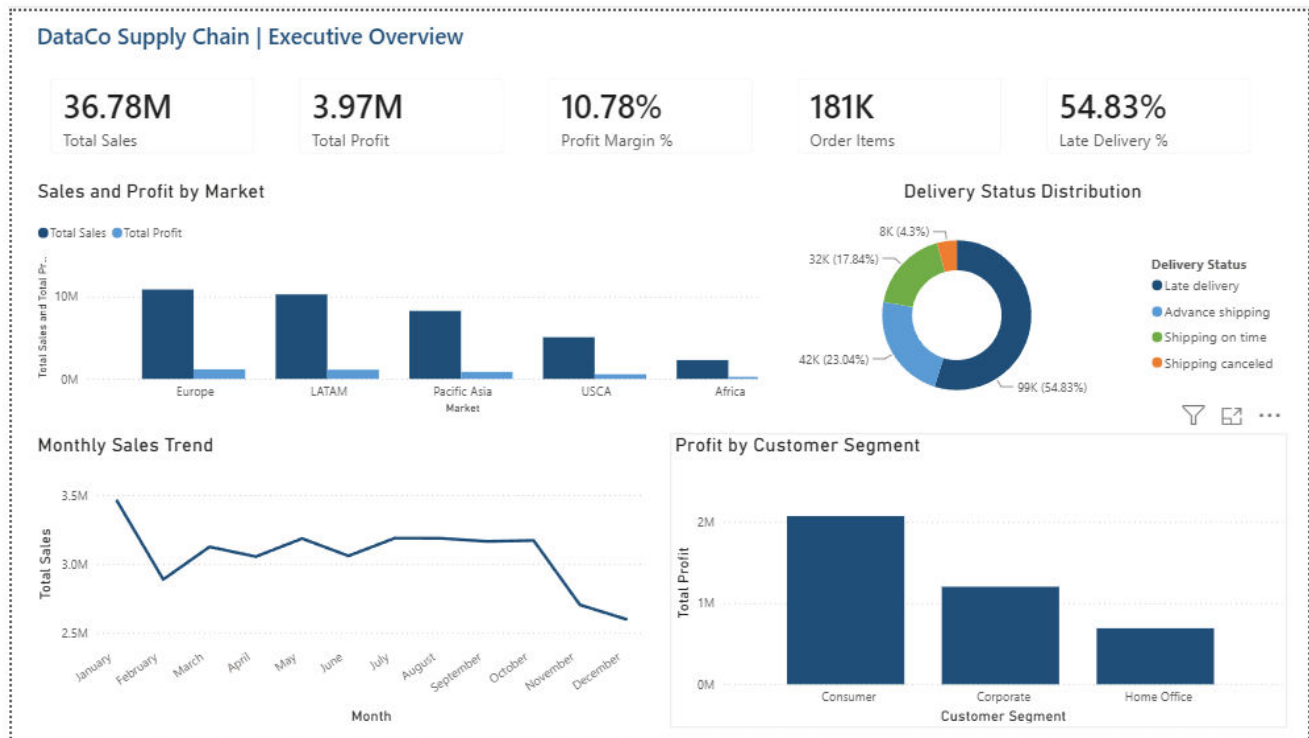


Figure 01 - Final Power BI Executive Overview.

Evidence from the executive view

- Sales: \$36.78M; profit: \$3.97M; profit margin: 10.78%.
- Order-item late delivery: 54.83%.
- Europe and LATAM carry the largest sales volumes among the displayed markets.
- Consumer is the largest customer segment by sales and profit.

Business implication

Delivery exposure is large enough to justify operational risk management, while the commercial dashboard shows where revenue and profit concentration make service reliability financially relevant.

05 Delivery and operational risk

Risk varies materially by shipping mode, while geographic differences are comparatively tighter at market level.

The strongest operational separation appears across shipping modes. The executed notebook records a 95.32% observed late-delivery rate for First Class, 76.63% for Second Class, 45.74% for Same Day, and 38.07% for Standard Class. These are historical descriptive rates and are not hard-coded rules in the ML application.

The final dashboard also shows the overall planning-versus-execution gap: average scheduled shipping is 2.93 days compared with 3.50 actual shipping days. The regional view highlights Central Africa, South Asia, and East Africa among the highest late-delivery risk concentrations shown in the dashboard.



Figure 02 - Final Power BI Delivery & Operational Risk.

Operational reading

EVIDENCE	INTERPRETATION
First Class: 95.32% historical late-delivery rate	High observed exposure warrants service-design and execution review.
Second Class: 76.63%	Material risk remains outside the highest-risk mode.
Standard Class: 38.07%	Lower observed risk does not remove the need for order-level screening.
2.93 scheduled vs 3.50 actual days	Planning and execution should be evaluated together.

06 Commercial performance

Service performance matters because it sits inside the economics of demand, margin, and customer value.

The commercial analysis shows Europe as the highest-sales market, Fishing as the leading sales category, and Consumer as the largest customer segment. Profitability is also sensitive to discount level: average profit declines from 23.94 in the 0-5% discount band to 20.14 in the 15-20% band.

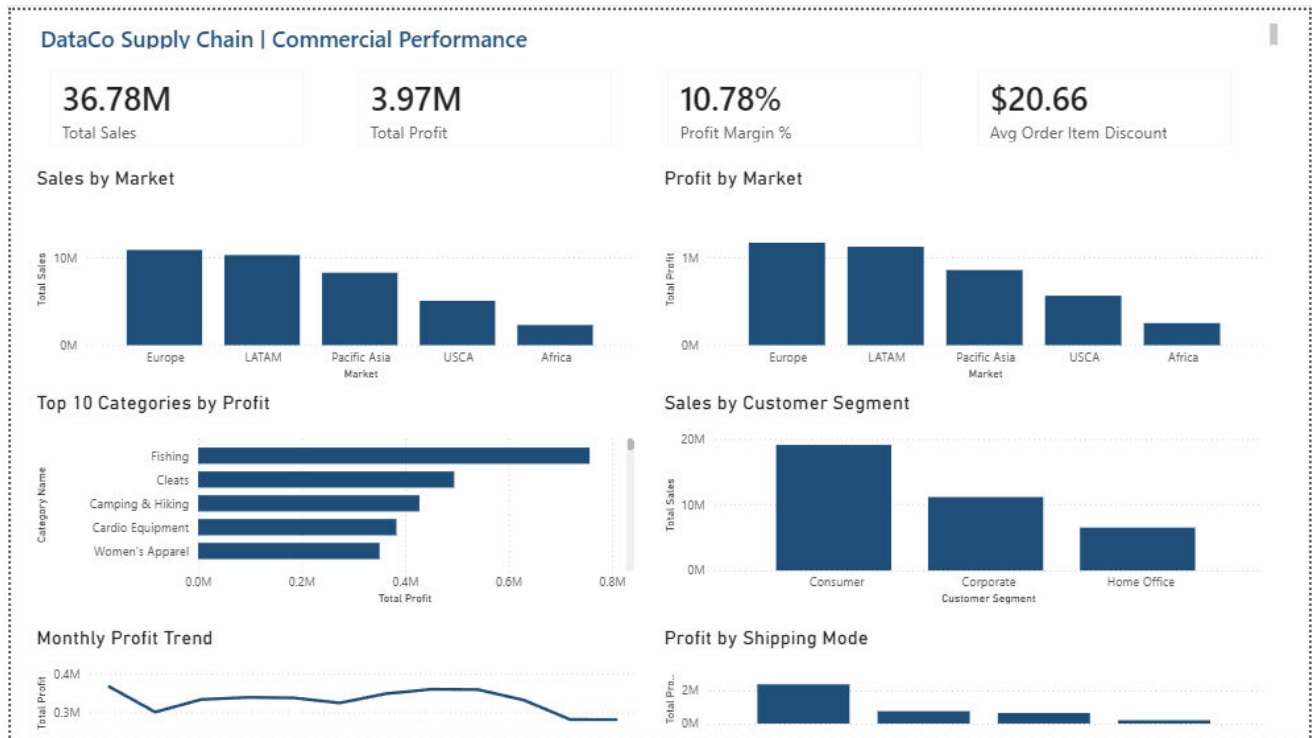


Figure 03 - Final Power BI Commercial Performance.

Commercial implications

- Revenue concentration should be evaluated together with profit, not sales volume alone.
- Discounting is associated with lower average profit across the analyzed discount bands.
- Shipping-mode profitability adds a service dimension to commercial decisions.

Business implication

Commercial discipline should combine market, category, segment, discount, and shipping analysis so that service decisions do not optimize revenue while weakening margin.

07 Predictive model development

The model is designed to screen late-delivery risk before shipment execution using order-time information.

The target is late-delivery status, encoded as a binary outcome. The final notebook separates the unseen test set before fitting preprocessing or models, uses a stratified 80/20 split, and applies the same preprocessing inside each candidate pipeline.

A baseline classifier establishes a reference accuracy of 54.83%. Random Forest and Gradient Boosting are then evaluated on the same holdout set. Gradient Boosting is selected because it provides the strongest precision among the evaluated candidates and produces probability estimates required for risk ranking and ROC-AUC.

Model comparison

METRIC	BASELINE	RANDOM FOREST	GRADIENT BOOSTING
Accuracy	0.55	0.70	0.70
Precision	-	0.83	0.84
Recall	-	0.56	0.55
F1	-	0.67	0.67
ROC-AUC	-	-	0.74

	Baseline	Random Forest	Gradient Boosting
Accuracy	0.55	0.70	0.70
Precision	NaN	0.83	0.84
Recall	NaN	0.56	0.55
F1	NaN	0.67	0.67
ROC-AUC	NaN	NaN	0.74

Figure 04 - Executed model comparison from the final notebook.

Model-selection rationale

- Gradient Boosting provides the highest observed precision among the candidate models.
- Probability output enables risk ranking instead of a binary prediction alone.
- The model is used as a screening layer; it is not presented as an autonomous operational decision system.

08 Model performance and interpretation

The evaluation is reported in full so that model capability and model limitations are visible together.

69.7% ACCURACY	84.3% PRECISION	55.0% RECALL
66.5% F1 SCORE	74.2% ROC-AUC	69.5% 5-FOLD CV ACCURACY

What the metrics mean operationally

METRIC	OBSERVED RESULT	BUSINESS READING
Precision	84.3%	A high share of orders flagged as late are actually late in the holdout evaluation; this supports targeted review.
Recall	55.0%	The model does not identify every late order; some late cases will remain outside the flagged set.
ROC-AUC	74.2%	ROC-AUC of 74.2% indicates moderate discrimination in the evaluated holdout set.
Cross-validation accuracy	69.5%	Performance is broadly consistent across the five validation folds.

Interpretation boundary

The evaluation describes performance on the executed dataset split. It does not guarantee production performance under future data, changing processes, or different operating conditions.

Feature behavior

The final notebook reports the highest feature-importance contributions from Days for shipment (scheduled) and Order Item Discount. Feature importance describes model behavior within this fitted model; it does not establish causality.

09 From model output to an operational action

The deployment layer converts probability into a review signal, recommendation, and financial context.

The Streamlit application loads the saved preprocessing-plus-model pipeline, validates the saved feature schema, accepts the nine order-time inputs, and returns both class prediction and probability. A presentation-level risk band is then applied to the probability: Low Risk below 40%, Medium Risk from 40% to 69.9%, and High Risk at 70% or above. The notebook financial scenario uses the model's binary holdout predictions and is separate from the Streamlit presentation-level 70% High Risk band.

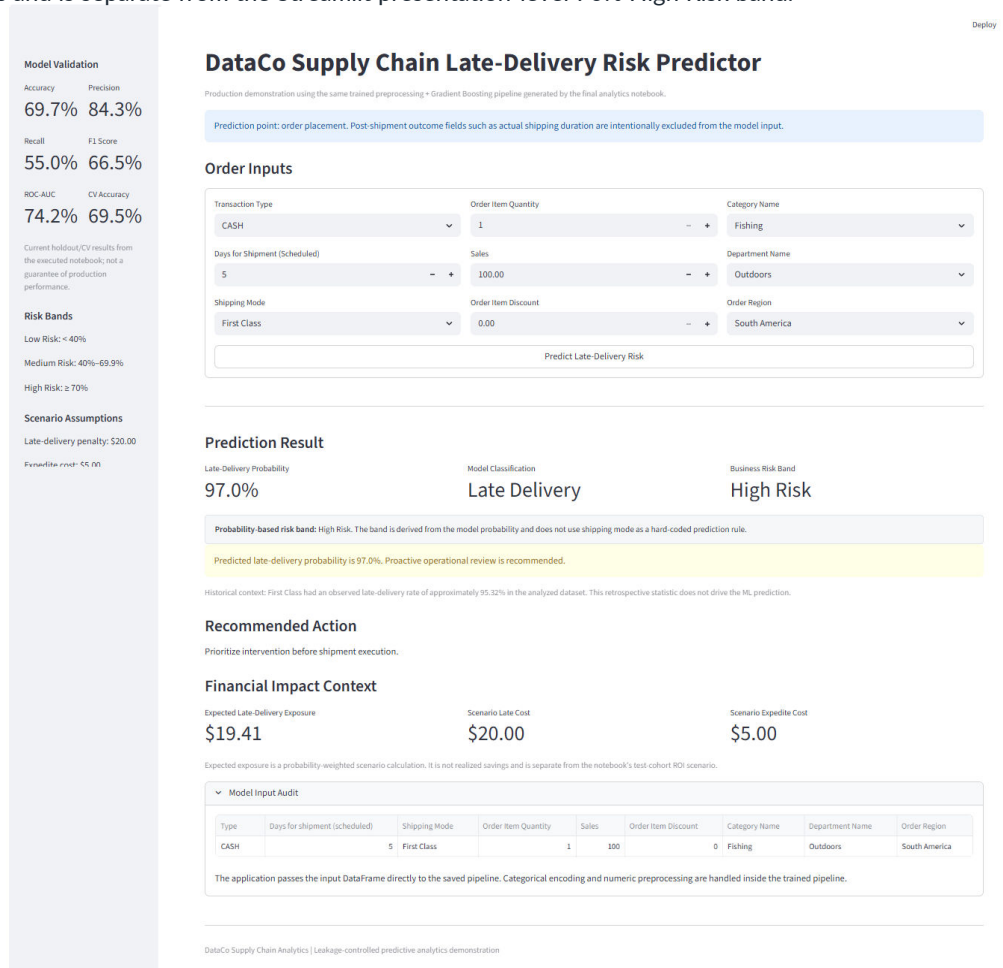


Figure 05 - Final Streamlit high-risk prediction demonstration.

Example order

97.0% predicted late-delivery probability → High Risk → Proactive intervention recommended → \$19.41 probability-weighted expected exposure under the \$20 scenario penalty.

Deployment controls

- The saved preprocessing is applied inside the trained pipeline rather than recreated separately in the application.
- The application validates the saved feature schema before prediction.
- Historical shipping-mode rates are shown as context and do not drive the prediction.
- Financial exposure is scenario context and is explicitly separated from the notebook test-cohort ROI scenario.

10 Financial impact scenario

The model is connected to a transparent scenario framework rather than an unsupported savings claim.

The executed notebook evaluates the model on 36,104 test orders, including 19,796 late orders. The financial scenario uses the model's binary holdout predictions to determine predicted interventions and missed late orders. Using a \$20.00 late-delivery penalty and a \$5.00 expedite cost, the modeled loss without ML is \$395,920 and the scenario cost with ML is \$242,840.

\$395,920 LOSS WITHOUT ML	\$242,840 SCENARIO COST WITH ML	\$153,080 ESTIMATED NET SCENARIO SAVINGS
38.66% SCENARIO ROI	\$4.24 ESTIMATED SAVINGS PER TEST ORDER	\$20.00 / \$5.00 PENALTY / EXPEDITE ASSUMPTIONS

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Test Orders          36,104.00
Late Orders          19,796.00
Loss Without ML      395,920.00
Scenario Cost With ML 242,840.00
Estimated Net Savings 153,080.00
Estimated ROI (%)    38.66
Estimated Savings per Test Order 4.24
dtype: float64
Scenario assumptions:
- Late delivery cost: $20
- Expedite cost: $5
These figures represent an assumption-based scenario, not realized financial savings.
  
```

Figure 06 - Executed financial impact scenario from the final notebook.

Scenario interpretation

ITEM	VALUE	MEANING
Loss without ML	\$395,920	Modeled late-delivery cost if every late order incurs the stated penalty.
Scenario cost with ML	\$242,840	Modeled expedite cost for predicted late orders plus penalty for missed late orders.
Estimated net scenario savings	\$153,080	Difference between the two scenario costs.
Scenario ROI	38.66%	Net scenario savings divided by modeled loss without ML.

Financial boundary

These figures represent an assumption-based scenario, not realized financial savings. Actual value would depend on intervention effectiveness, operating costs, adoption, process changes, and live production performance.

11 Recommendations and operating model

The recommendations connect the analytical findings to specific operating actions.



Figure 07 - Final Power BI Executive Recommendations.

Delivery risk

Action: Prioritize operational review where late-delivery exposure is concentrated by shipping mode and region.

Shipping strategy

Action: Review service design and execution for high-risk shipping modes; use historical rates as evidence, not as hard-coded ML rules.

Commercial discipline

Action: Evaluate market, category, segment, discount, and shipping performance together to protect profitable demand.

Predictive operations

Action: Use the probability-based model as a pre-shipment screening layer and route high-risk orders for review.

Financial prioritization

Action: Use the explicit \$20 penalty and \$5 expedite assumptions when evaluating intervention scenarios, then validate those assumptions with actual operating data.

Proposed operating loop

Order → Score → Prioritize → Review → Intervene → Measure

12 Business value

The value of the work comes from connecting visibility, prioritization, prediction, action, and financial context.

- | | |
|----|--|
| 01 | Visibility
54.83% late-delivery exposure becomes a quantified operating baseline. |
| 02 | Prioritization
Shipping-mode and regional analysis show where operational review can be concentrated. |
| 03 | Commercial protection
Sales, profit, segment, category, discount, and shipping analysis put service performance in its financial context. |
| 04 | Predictive screening
Order-time features are converted into a probability-based late-delivery risk signal. |
| 05 | Operational action
The Streamlit layer converts probability into a risk band and recommended review action. |
| 06 | Financial context
The notebook scenario estimates \$153,080 net scenario savings under explicit assumptions. |

What changes operationally

The completed workflow changes the use of the dataset from retrospective reporting to an earlier intervention model. Teams can see where delivery exposure is concentrated, evaluate commercial trade-offs, and screen individual orders before shipment using a consistent saved pipeline.

Quantified scenario value

\$153,080 estimated net scenario savings | 38.66% scenario ROI | \$4.24 estimated savings per test order.

The financial figures remain scenario-based. Their value is to make intervention economics explicit and testable, not to imply that the modeled savings have already been realized.

13 Technical integrity, artifacts, and limitations

The final package is designed so that analytical conclusions, deployment behavior, and financial assumptions remain traceable.

ARTIFACT	FINAL FILE	ROLE
Analytics and ML	DataCo_Supply_Chain_Analytics_and_ML_FINAL.ipynb	Data audit, business analysis, predictive modeling, evaluation, scenario analysis, artifact handoff.
Power BI	DataCo_Supply_Chain_Flagship_v4.pbix	Executive performance, operational risk, commercial analysis, and recommendations.
Streamlit	app.py	Pre-shipment prediction, probability-based risk band, recommendation, and scenario exposure.
Model pipeline	supply_chain_model.pkl	Saved preprocessing and Gradient Boosting model.
Feature schema	model_features.pkl	Validated model input schema.
Model metrics	model_metrics.json	Saved evaluation metrics used by the application.
Scenario configuration	business_config.json	Late-delivery penalty and expedite-cost assumptions.
Cleaned data	DataCoSupplyChainDataset.csv	Analytical dataset used by the project.

Limitations and controls

- Historical relationships are descriptive and are not presented as causal effects.
- The model is evaluated on the holdout set and cross-validation folds; future production performance may differ.
- Recall of 55.0% means some late-delivery cases will remain unidentified.
- The risk bands are presentation-level thresholds applied to model probability.
- The financial scenario depends on explicit assumptions and should be validated against observed intervention costs and outcomes.
- Post-shipment outcomes are excluded from predictive inputs to preserve the order-placement prediction boundary.

Integrity statement

The report separates observed historical evidence, model evaluation results, deployment behavior, and scenario assumptions. No scenario estimate is presented as realized business performance.

DATA ANALYTICS CASE STUDY

DataCo Supply Chain Analytics & Predictive ML

From data to operational decision support.

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Project evidence

Analytics notebook | Power BI dashboard | Saved ML pipeline | Streamlit deployment | Cleaned analytical dataset

This case study is designed to stand on its own: the business problem, analytical evidence, predictive capability, operational workflow, financial scenario, recommendations, and limitations are documented within the report.